

# Ear-canal Characterisation for Optimum In-Ear Headset User Experience

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## ABSTRACT

Users listening to music, making phone calls or listening to the environment they are in (i.e. situational awareness) are having a sub-optimal user experience just because the in-ear headsets they are using are not optimised for each user but rather optimised for a general population (i.e. population average). The proposed method is one way of alleviating this problem by compensating for the users' ear-canal shapes and size difference and including a correction EQqualiser (EQ) into the signal processing chain, for best and most compelling user experience.

## CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**.

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## 1 INTRODUCTION

In normal listening with open ears, the incident sound waves are modified by the listener's body, head, and outer ear. When sounds are recorded with a conventional microphone situated away from the body and played back through headphones, the modifications caused by the body of the listener are lost. However, when microphones are placed binaurally near the ear canal entrances, the majority of these modifications are preserved.

The main difference when using in-ear headphones compared to the listening with open ears is that the headphones occlude the ear canals completely. An open ear canal acts as a quarter-wavelength resonator, (i.e. like a tube with one end being open and the other end closed by the ear drum). The first quarter-wavelength resonance of the open ear canal occurs at approximately 2–4 kHz. When an in-ear headphone blocks the ear canal, it becomes a half-wavelength resonator. The closed ear canal does not only create

a new half-wavelength resonance, but it also cancels the quarter-wave resonance created by the open ear canal, which people are accustomed to hearing. The first half-wavelength resonance occurs at approximately 5–10 kHz, depending on the length of the ear canal and the fitting of the headphone.

These resonance behaviors need to be taken into account when using in-ear headphones. The basic idea is to make the in-ear headphones sound natural (i.e. as if one would listen with open ears). The headphones need to cancel the unnatural half-wavelength resonance and re-create the missing quarter-wavelength resonance in order to sound natural. This has to be done per user (since ear-canal vary in shape and size amongst users) for a more compelling listening experience.

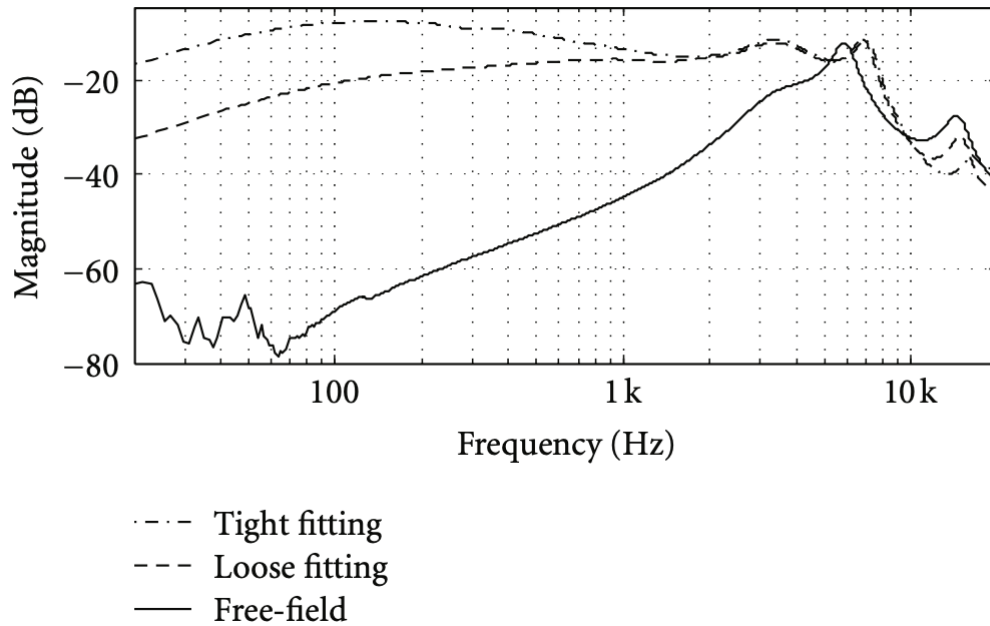
When listening to loud-speakers in a room, sound pressure waves are radiated to the whole room the listener is in. When using headphones, especially in-ear headphones, the volume of the space to which the wave is confined is extremely small compared to the volume of a room. The cavity between the in-ear headphone and the ear drum is about 1 cm<sup>3</sup>. With such a small cavity it is easy to produce high sound pressure levels. At low and middle frequencies, where the wavelength is large compared to the earphone driver, the pressure inside the ear canal is in phase with the volume displacement of the transducer membrane and its amplitude is proportional to it [1]. Therefore, the pressure chamber principle enhances the low and middle frequencies, as seen in Figure 1.

In principle, to achieve the pressure chamber effect, the headphone should be tightly fit so that there would be no leaks, but in reality small leaks do not interfere much in mid to high frequencies compared to its effect on low frequencies [1]. Figure 1 shows the frequency response of an in-ear headphone measured in free-field and in an ear canal simulator with tight fitting and with loose fitting. As can be seen, when the headphone is fit loosely into the ear canal simulator, the level of low frequencies decreases.

## 2 PROPOSED SOLUTION

In order to provide a more user personalised in-ear headphones experience, we need to capture the shape and size ear-canal variation (i.e. capture the user's ear-canal transfer function), and then design an user centric equaliser which would result in best personlaised user experience. Figure 2 shows an example of measuring the effect of an open-ear versus and occluded ear (i.e. when a user is wearing an in-ear device), such as the test setup in Figure 3a. It is very evident in Figure 2 how the quarter wavelength disappears and a half wavelength appear (when in-ear device is worn by the user).

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**Figure 1: Frequency response of a headphone measured in free- field and in an ear canal simulator with tight and loose fitting (with 1/12 octave smoothing) [2]**

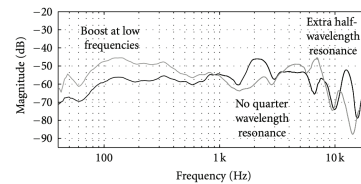
In addition to the above effect, the original quarter wavelength position differs between users due to differing shape and size of each user’s ear-canal. There maybe even subtle difference between left and right of the same user. In the proposed setup, when a user first puts the in-ear device on during the day (and potentially every time he puts it back on), ear-canal characterisation will take place which will generate an EQ filter for that session/fit.

Before detailing how such ear-canal characterisation is conducted and the types we consider, we need to highlight the fact that in the system setup of Figure 3a, the ear-canal characterisation would be optimum at the inner microphone position. Ideally, we want this equalisation to happen at the ear-drum for best user experience, therefore we need to find the transfer function  $H$ , as shown in Figure 3a, which virtually positions the inner microphone at the ear-drum position. The  $H$  transfer function maybe found offline on an artificial head (such as B&K artificial head) for each new in-ear model and included into the EQ filter generation as performed in the ear-canal characterisation sub block.

In the proposed system, we consider two characterisation approaches detailed as follows.

## 2.1 Fixed Ear-Canal Characterisation

With reference to Figure 3a, in the fixed ear-canal characterisation, a test signal is played through the speaker into the ear-canal cavity while the user is wearing the in-ear device. There are a number of test signals that could be used in this setup (such as a log-chirp



**Figure 2: Transfer functions from an external sound source to an ear canal: black curve depicts an open ear case and gray curve is measured when the sound travels through an unequalised in-ear headset into the ear canal. [3]**

or a Maximum Length Sequenc MLS), but in the setup shown in Figure 3a, a log-chirp is used. The transfer function from speaker to microphone is shown in Figure 2, as recorded by the microphone. Speaker to microphone transfer fuction is first translated to the ear-drum position through transfer function  $H$ . Then the equalisation filter generation would remove the effect of user putting the in-ear device and would look like Figure 3b. The EQ is then included in the signal processing chain. The effect of which is removing the low-frequency boost as well as the half wavelength artifacts of wearing the in-ear device and re-introduces the quarter wavelength for natural hearing prior to wearing the in-ear device.

## 2.2 Adaptive Ear-Canal Characterisation

The fixed method works well but it is a bit intrusive and requires a test signal to be played through the speaker every time a new user fit takes place (i.e. putting the in-ear device on after a period of it being

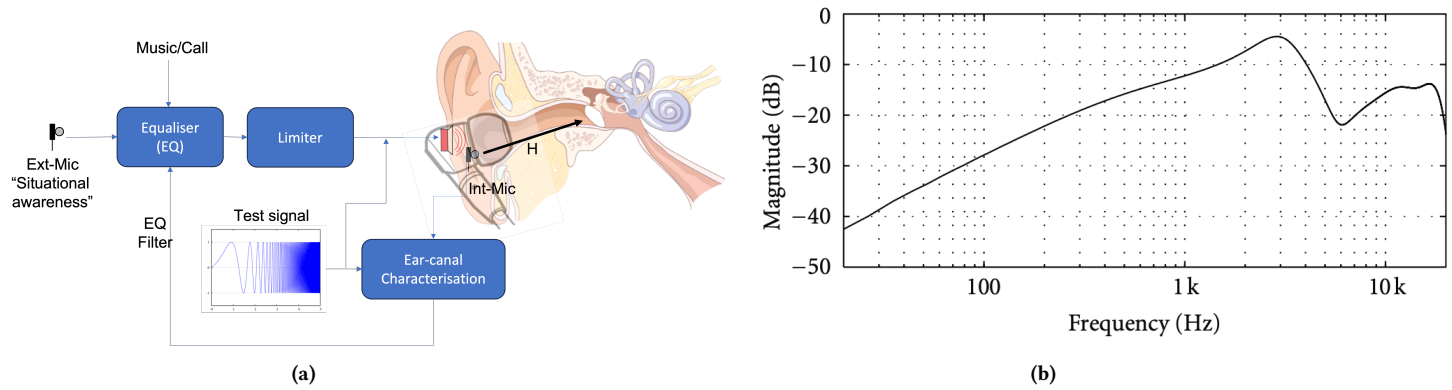


Figure 3: (a) In-ear system overview. (b) A generic equalization curve [3].

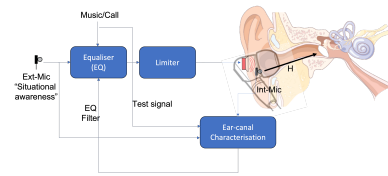


Figure 4: Adaptive ear-canal characterisation

not worn). A less intrusive method would be to use the content used by the user (whether it is music playback, situational awareness, or call) to adapt the EQ filter continuously while including the  $H$  transfer function (shown in Figure 3a) and the removal of the half wavelength and reinstating the quarter wavelength, as shown in Figure 4. All of these requirements could be included in the adaptive block and the filter would eventually converge to an optimum EQ for that fit. The adaptive method would continuously adapt and

the EQ filter gets updated at a much slower rate to remove the oscillation effects.

### 3 CONCLUDING REMARKS

In conclusion, users have differing ear-canal shapes and sizes. While in the past it was acceptable to have one EQ filter that fitted everyone, it did not result in most compelling user experience. The proposed method provides a solution that is user centric and would lead to a more personalised and compelling in-ear headset user experience whether it is used for music playback, situational awareness or call.

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